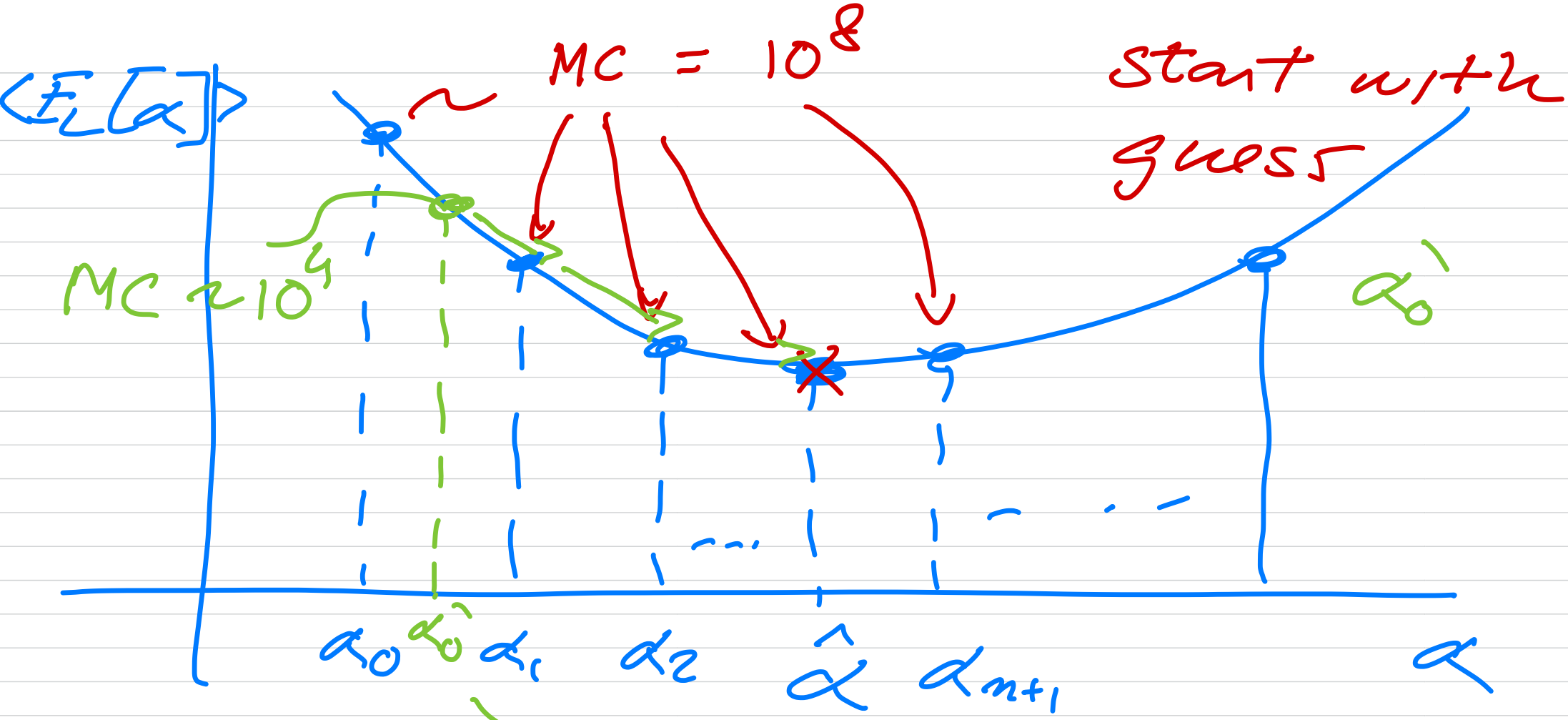
F454411/9911 Feb 27

$$E[\vec{\alpha}] = \frac{\int \psi_T^*(\vec{R};\vec{\alpha})\, H(\vec{R};\vec{\sigma})\, \psi_T(\vec{R};\vec{\alpha}) \times d\vec{R}}{\int |\psi_T(\vec{R};\vec{\alpha})|^2}$$

$$\mathbb{E}[E_L(\alpha)] = \int P(\vec{R};\vec{\alpha})\, E_L(\vec{R};\vec{\alpha})\, d\vec{R}$$



$$\vec{\nabla}_\alpha \mathbb{E}[E_L(\alpha)]$$

$$\vec{\alpha} = \{\alpha\}$$

Newton's method

$$\mathbb{E}[E_L(\alpha)] = E(\alpha)$$

$$\vec{\alpha} \to \alpha$$

$$\vec{\nabla}_{\vec{\alpha}} E(\vec{\alpha}) = \vec{g}(\vec{\alpha})$$

$$\vec{g}(\vec{\alpha}) \to g(\alpha)$$

$$\alpha_{opt} = \hat{\alpha}$$

Prepare for Taylor expansion

Expand around $\hat{\alpha} - \alpha_n$

$$g(\alpha) \to g(\alpha_m) = \left.\frac{dE(\alpha)}{d\alpha}\right|_{\alpha=\alpha_m}$$

$$= g_m$$

$$\nabla_\alpha^2 E(\alpha) \to \nabla_\alpha^2 E(\alpha_m) = A_m$$

Taylor expand around $\hat{\alpha}$

$$E(\hat{\alpha}) = E(\alpha_m) + \underbrace{(\hat{\alpha}-\alpha_m)}_{b_m} \cdot g_m$$

$$+ \frac{1}{2} A_m (\hat{\alpha}-\alpha_m)^2$$

$$+ O\left((\hat{\alpha}-\alpha_m)^3\right)$$

$$E(\hat{Q}) = E_m + g_m b_m + \frac{1}{2} A_m b_m^2 + O(b_m^3)$$

(as vectors

$$E(\hat{Q}) = E_m + \vec{g}_m^T \vec{b}_m + \frac{1}{2} \vec{b}_m^T A_m \vec{b}_m + O(\vec{b}_m^3)$$

$$\vec{\alpha} = \{ \alpha_0, \alpha_1 \}$$

$$\vec{\alpha}_n = \{ \alpha_0^{(n)}, \alpha_1^{(n)} \}$$

$$A_n = \begin{bmatrix} \left.\dfrac{\partial^2 E}{\partial \alpha_0^2}\right|_{\alpha_0 = \alpha_0^{(n)}} & \left.\dfrac{\partial^2 E}{\partial \alpha_0 \partial \alpha_1}\right|_{\vec{\alpha} = \vec{\alpha}^{(n)}} \\ \left.\dfrac{\partial^2 E}{\partial \alpha_1 \partial \alpha_0}\right|_{\vec{\alpha} = \vec{\alpha}^{(n)}} & \dfrac{\partial^2 E}{\partial \alpha_1^2} \end{bmatrix}$$

Neglect $O(b_n^3)$

$$E(\vec{\alpha}) = E_n + \vec{g}_n^T \vec{b}_n + \frac{1}{2} \vec{b}_n^T A_n \vec{b}_n$$

$$\left(f(\vec{x}) = C + \vec{g}^T \vec{x} + \frac{1}{2} \vec{x}^T A x \right)$$

$$\frac{\partial f}{\partial x} = 0 = A\vec{x} + \vec{g} \Rightarrow$$

$$\vec{x} = A^{-1} \vec{g}$$

simple example

$$f(x_1, x_2) = x_1^2 + x_1 x_2 + 10 x_2^2 - 5x_1$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$f(x_1, x_2) = \frac{1}{2} x^T \begin{bmatrix} 2 & 1 \\ 1 & 20 \end{bmatrix} x$$

$$- \begin{bmatrix} 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\frac{\partial f}{\partial x_1} = 0 = 2x_1 + x_2 - 5$$

$$\frac{\partial f}{\partial x_2} = 0 = x_1 + 20x_2 - 3$$

Guess for iteration

$$x_0 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

what we need is

$$\vec{\nabla}_\alpha E[\hat{\alpha}] \wedge \hat{\alpha} = \alpha_{n+1}$$

$$\vec{\nabla}_\alpha E(\alpha_{n+1}) = 0 = \underbrace{\vec{g}_n}_{\vec{\nabla}_\alpha E(\vec{\alpha}_n)} + A_n \vec{b}_n$$

$$\vec{b}_n = \vec{\alpha}_{n+1} - \vec{\alpha}_n$$

$$\vec{\alpha}_{n+1} - \vec{\alpha}_n = -A_n^{-1} \vec{g}_n$$

$$\boxed{\vec{\alpha}_{n+1} = \vec{\alpha}_n - A_n^{-1} \vec{g}_n}$$

Newton-Raphson's iterative method.

solve iteratively starting with a guess $\vec{\alpha}_0$ and repeat till

$$\|\vec{\alpha}_{m+1} - \vec{\alpha}_m\|_2 \leq \varepsilon \sim 10^{-5}$$

$$\|\vec{x}\|_2 = \left(\sum_{i=1}^{n} x_i^2\right)^{1/2}$$

$$\|\vec{x}\|_2^2 = \sum_{i=1}^{n} x_i^2$$

$$\|\vec{x}\|_1 = \sum_{i=1}^{n} |x_i|$$

we want to avoid to evaluate $A_n \Rightarrow \underline{\gamma_n}$

learning rate

$$\vec{\alpha}_{n+1} = \vec{\alpha}_n - \gamma_n \vec{g}_n$$

$$\left(\vec{\alpha} \leftarrow \vec{\alpha} - \gamma_n \vec{g}_n \right)$$

$\gamma_n = \gamma = \text{constant} \Rightarrow$ plain gradient descent.

Can we find an optimal γ_n ?

$$\hat{\vec{\alpha}} = \vec{\alpha}_{n+1} = \vec{\alpha}_n - \gamma_n \vec{g}_n$$

Expand

$$E(\vec{\alpha}_n - \gamma_n \vec{g}_n) \simeq$$

$$E_n - \gamma_n \vec{g}_n^T \vec{g}_n + \frac{1}{2} \gamma_n^2 \vec{g}_n^T A_n \vec{g}_n$$

$$\frac{dE}{d\gamma_m} = 0$$

$$\gamma_m = \frac{\vec{g}_m^T \vec{g}_m}{\vec{g}_m^T A_m \vec{g}_m}$$

steepest descent

Matrix with $D^2 E$

We don't want evaluate A_n $\Longrightarrow$

1) Quasi-Newton

a) Broyden's method

b) Broyden-Fletcher-Goldfarb-Shanno BFGS

c) Powell

⋮

2) GD + momentum

SDG — " —

ADAM5

ADAgrad

RMSprop

when $\vec{\alpha}$ is large

$\vec{\alpha} \in \mathbb{R}^p$

$$\vec{\nabla}_{\vec{\alpha}} E(\vec{\alpha}) = ?$$

$$E_L(x;\vec{\alpha}) = \frac{1}{\psi_T(x;\vec{\alpha})} H(x)\,\psi_T(x;\vec{\alpha})$$

$$\psi_T(x;\vec{\alpha}) = \psi_\alpha(x)$$

$$P_\alpha(x) = \frac{|\psi_\alpha(x)|^2}{N_\alpha}$$

$$N_\alpha = \int dx\, |\psi_\alpha(x)|^2$$

$$\int dx\, P_\alpha(x) = \underline{1}$$

$$\langle E_L(\vec{\alpha})\rangle \to E(\alpha) =$$

$$\int dx\, P_\alpha(x)\, E_L(x;\alpha) = \langle E_L \rangle$$

start from

$$E(\alpha) = \frac{\int dx\, \psi_\alpha^*(x)\, H\, \psi_\alpha(x)}{\int dx\, |\psi_\alpha(x)|^2}$$

assume ψ_α real

$$A(\alpha) = \int dx\, \psi_\alpha H \psi_\alpha$$

$$B(\alpha) = \int dx\, \psi_\alpha^2(x)$$

$$E = A/B$$

$$\partial_\alpha E = \frac{(\partial_\alpha A) B - A(\partial_\alpha B)}{B^2}$$

$$\partial_\alpha B = \partial_\alpha \int dx\, \psi_\alpha^2 = \int dx\, 2\psi_\alpha \times \partial_\alpha \psi_\alpha$$

$$\partial_\alpha = \frac{\partial}{\partial \alpha}$$

$$\partial_\alpha A = \int dx \left[(\partial_\alpha \psi_\alpha) H \psi_\alpha + \psi_\alpha H (\partial_\alpha \psi_\alpha) \right]$$

H is Hermitian.

$$\int dx\, \psi_\alpha H (\partial_\alpha \psi_\alpha) =$$

$$\int dx (\partial_\alpha \psi_\alpha) H \psi_\alpha$$

$$\partial_\alpha A = 2 \int dx (\partial_\alpha \psi_\alpha) H \psi_\alpha$$

$$\partial_\alpha E = 2\left(\int dx\,(\partial_\alpha \psi_\alpha)\,H\,\psi_\alpha\right)B$$

$$(A = EB)$$

$$- (EB)\left(\int dx\, 2\psi_\alpha\,\partial_\alpha \psi_\alpha\right)$$

$$\overline{\qquad\qquad\qquad

Divide and multiply by ψ_α

$$\frac{\int dx\,(\partial_\alpha \psi_\alpha)\, H \psi_\alpha}{\int dx\, \psi_\alpha^2}$$

$\partial_\alpha \ln \psi_\alpha$

$$= \frac{\int dx\, \psi_\alpha^2 \left(\frac{\partial_\alpha \psi_\alpha}{\psi_\alpha}\right) \underbrace{\left(\frac{H\psi_\alpha}{\psi_\alpha}\right)}_{E_L}}{\int \psi_\alpha^2\, dx}$$

$$= \int P_\alpha(x) \left[\partial_\alpha \ln \psi_\alpha \, E_L \right] dx$$

$$= \langle \partial_\alpha \ln \psi_\alpha \, E_L \rangle$$

collecting

$$\partial_\alpha E = 2 \Big[\langle \partial_\alpha \ln \psi_\alpha \, E_L(\alpha) \rangle$$

$$- \langle E_L(\alpha) \rangle \langle \partial_\alpha \ln \psi_\alpha \rangle \Big]$$

$$\partial_\alpha \ln \psi_\alpha(x) = \frac{\partial_\alpha \psi_\alpha(x)}{\psi_\alpha(x)}$$

$$= O_\alpha(x)$$

$$\partial_\alpha \overset{\langle E_L(\alpha)\rangle}{E} = 2\left(\langle O_\alpha E_L\rangle - \langle O_\alpha\rangle\langle E_L\rangle\right)$$

$$= 2\,\mathrm{cov}(O_\alpha, E_L)$$

$$\frac{\partial_\alpha \psi_\alpha(x)}{\psi_\alpha} = \partial_\alpha \ln \psi_\alpha(x)$$

$$\psi_\alpha(x) \sim e^{-\frac{1}{2}\alpha^2 x^2}$$

$$\psi_{\vec{\alpha}}(x) \sim \prod_{i=1}^{N} e^{-\frac{1}{2}\alpha_0^2 x_i^2}$$

$$\times \prod_{i<j}^{N} f(x_{ij})$$

$$f(x_{ij}) \sim e^{\pm \alpha_1 x_{ij}}$$

Lecture note example ψ_α

$$\psi_{\alpha,\beta}(\vec{x}_1, \vec{x}_2) = \underbrace{e^{-\frac{1}{2}\alpha^2(x_1^2+x_2^2)}}_{\text{one body part}} \; \underbrace{e^{\frac{x_{ij}}{1+\beta x_{ij}}}}_{\text{correlation Jastrow}}$$

$$x_{ij} = \sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$$

Exact energy = 3 a.u.

$\hbar\omega = 1$ a.u.